ActInf Textbook Group ~ Cohort 7 ~ Session 20 (Chapter 8) ~ 3/20/2025

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Welcome everyone to the Active Inference Institute textbook reading group. This is cohort 7 continuing on with chapter 8 of the textbook. We already know that we are in the second half of the textbook, meaning we're much more focused on practice uh rather than theory uh which we had in the first part. In chapter 2, we were introduced in chapter six, excuse me, in part two, we were introduced in chapter six to a recipe for building active inference models. So we're familiar with all of uh you know the core fundamentals including what is your system of interest or your agent or your generative model that you are looking at and then what is on the other hand uh to use Mark Solmes's words uh just as we identified a system of interest we have to figure out our not system that is what is not the system of interest what is the environment or other variables uh that you know exist in the world but are not um part and parcel with our agent. Right? So that's that means identifying what is our generative uh process or the environment or the not system. And so we are familiar with a lot of the core components. We look at things like hidden states, observations and uh actions or policies which are sequences of actions. And then we identify the variables of the models, the different parameters. And from there we just build upward and we look at what kind of model is necessary for understanding our particular system of interest. And uh so we looked at chapter 7 which had to do with discrete state space models and that covered largely the hidden Markov model in the musician example and then built up to the POMDP or partially observable Markov decision-m process uh which is sort of the exemplar discrete state space model and that uh was illustrated through use of the T-mase example mouse choosing left or So we know already with the discrete state space that we if we followed up to the end of chapter 7, we can build models where there are discrete actions being taken like the mouse chooses the left or the right arm of the maze or it chooses to visit the informative cue along the way. But these ultimately amount to three different actions per time step. Um they are not a continuous state space. Right? So for those who are very familiar with anything from just general statistics linear regression up to machine learning and deep learning models that work with continuous state spaces often times we want to create models that can predict continuous values such as 37.99 as opposed to choose left in the teammates right and that also means being able to read continuous data such as from data streams IoT devices um financial data uh there's a plethora of different kinds of continuous uh data to look at. So that's what brings us here to chapter 8 looking at continuous time models. The these are arguably where active inference started. Um neuronal signals are usually measured continuously including from uh electronphilography or EEG uh devices as well as uh fMRI and looking at bolt signals as well as a variety of other physiological data including things like heart rate. So whenever we get into chapter eight, uh we turn to continuous state space models which the authors state are well suited for modeling physical fluctuations impinging on sensory receptors and for the continuous motion of the aectors, eg the muscles we used to change the world around us with model examples for the dynamical systems involved in motor control as well as the concept of generalized synchrony and finally constructing hybrid models for combining discrete and continuous variables at different levels which features at the end of the chapter. So largely we're introduced to notation for stochastic equations uh which invol are involved in continuous time models. Uh we see a lot of similar notation to what we've seen so far earlier on in the textbook. The authors largely use y to denote uh observations and x to denote hidden states. here. This is not terribly different, but we notice that we're also looking at and excuse me, let me go to this is technically the one for the generative model. Um, here we're also looking at a more continuous state space. We're looking at something like a kind of error term or or or flux uh or or stochasticity within an equation as well as we're seeing a derivative. So we're looking at uh states evolving over time as a function of hidden states um as well as well or rather what is the rate of change of x given x and some kind of static or otherwise variable with stochasticity. So from there we're introduced to other notions. These are all very kind of standard ideas that of course are absolutely not unique to active inference. This is active inference building upon a lot of previous work. And so we're looking at an attractor system, right? So, so this uh generative model in equation 8.3, we're introduced to this idea that V is a kind of attractor. So whenever X is too high, it tends to become its rate of change becomes negative as it heads back towards the value of the attractor. and vice versa. If X is too low tends to increase back towards the value of the attractor. And so that means as an attractor is a kind of attracting or fixed point. Um, I I I emphasize this despite its simplicity because when you think about it, we're oftentimes trying to model systems that, you know, whenever it comes to agents or looking at organisms or anything that needs to persist in some kind of way within an environment, whether it means an organism trying to take care of its homeostasis or it be a thermostat reacting to changes in temperature, we tend to be dealing with something like attractors. we we tend to deal with, you know, an attractor for me is that my body temperature tends to be roughly around 98 degrees Fahrenheit, as I use Fahrenheit in the United States. And uh so so we're looking at attractors in the sense of just where where do things tend around, right? Uh without the attractor, we can potentially completely disperse depending on how our model is set up. There's nothing to kind of guide us in our directionality. And whenever it comes to maintaining a bodily or physiological or other kind of system that requires maintenance at some homeostatic level or range, we can think of the homeostatic level or range as very much, you know, kind of an attractor or set of attractors uh in which the model tends towards to persist. Right? So, um we see a lot of these different kinds of dynamics play out. We're also introduced to precision, attention, and sensory attenuation. Many of these make sense to introduce here. Precision not and potentially attention. Uh they they could come in more deeply in earlier chapters and of course they do get mentioned in previous chapters, but essentially they make sense here because they tend to be viewed as continuous values. Not necessarily, but um so so they just relate much more to the material in this particular chapter with precision as a kind of sharpening of a probability distribution. So if precision is uh below zero or excuse me between zero and one. So what can happen is that your probability distribution can come become more uniform and vice versa if your precision is far above or even a little bit above one then the precision of your probability distribution is going to sharpen right so a general sense of this is that if I have a strong amount of trust in the observations I'm seeing that is I can clearly tell that the thing that I'm seeing in the distance I feel fully confident it is what it is and we kind of liken that to the idea of precision that means there's a very high probability that the thing I see in the distance is is what I think it is. The low near zero probability that it's going to be something else. Right? So we get this very sharp distribution where a lot of the the density is centered around one possibility. Right? Um similarly uh the other way around if precision is low we we see a much more flat broader distribution. I have no idea what that thing in the distance is and I'm not very clear on it. So precision in that sense is kind of like a sharpening or lessening dynamic. It can be related to synaptic gain uh in in neuronal dynamics. Um it it's it's seen as being very crucial for uh looking at things like confidence, right? Whether it be confidence in your your A matrix, your uh in terms of being able to uh interpret a situation or a B matrix in terms of like being able to look forward or back. And whenever we're looking at continuous time models where we're no longer in the world of A and B matrices, although some of the dynamics are certainly similar in the sense of temporality that we find in a B matrix and relating uh observations or data or Y to hidden states or higher order beliefs or X. So um we're also looking a lot at examples that use dynamical uh systems and so a good example is the generalized locker dynamics. This is kind of a classic ecological model that looks at uh relationships between plants, herbivores and car carnivores. The sort of common sensical notion of it is that the carnivores eat the herbivores who eat the plants and then the plants need enough time to regrow otherwise their population will completely die out. Um which of course because you can see the dependencies between these three different um organisms. They their their populations fluctuate over time as one proceeds to consume the other and yet they come back with reproduction and yet further if they were to to strongly diverge from these kinds of patterns. What we we would have is eventual extinction of at least one of the species which necessarily in this situation would lead to extinction of the rest of the remaining species. Right? So we see this kind of movement of them in kind of cyclical patterns nearly. And whenever we compare the herbivores and the plants or we compare the carnivores or the herbivores over the course of playing out this simulation over time, we see that there is a kind of attracting shape to the continuous state space that is their populations or their joint um populations. or the way of looking at them as they correlate with one another in different ways over time. Um, we see these shapes play out as well whenever we look at the Laurens system. And this is from the bird song example, which is where there are two birds who are singing with one another. And the idea is that they're going to kind of collaboratively finish a song where they take turns filling in space for the notes. I almost think of two guitarists playing solos or something and they're like trading off back and forth and yet they're still maintaining the same song. So there's a kind of synchronicity, the kind of rhythm that they need to maintain. Uh they need to stay on beat as it were. Um, and there's variation in what they play over time, right? They're not just both playing the same notes to where we get some very, you know, we don't settle in some clear equilibrium and stay there. There's a lot of variety going on and yet they have to stay in step. And um, ideally they are in fact playing the same song. So whenever we look at the Laurens system and we look at kind of what's going on in the state spaces here, we find attraction around two general areas of kind kind of like two spheres of orbit, right? Um as these two birds sing their songs over time and then they continue to explore this state space as they go. Right. So we see like or before excuse me let's go with this after learning they learn one another's patterns such that the state space they're exploring it tends to follow follow this line that is they're they're much more kind of synchronized and we notice relative to before learning where things were much more chaotic as they're still becoming attuned to one another's tunes. uh after learning not only is it more synchronized but we notice the axes have kind of increased right so in before learning they explore actually a significantly smaller area of the state space as their kind of correlated line between their actions you know spreads around after learning it's much more focused and they actually make it much further in different directions of the state space presumably they've learned to really well attune to one another while simultaneously being able to hit new notes that maybe weren't heard before. The notes presumably being part of the state space or identifiable therein. So a lot of this starts to look like when whenever we consider attractors but then we expand it to where there are multiple agents. It's interesting to see the different kinds of emergent phenomena that kind of arise out of the dynamics that play out between them and how they synchronize because in active inference we've often talked about um agents who sort of that their internal states you know presumably if the the agent is is is sort of optimal enough for the environment that they're in and find ways to adapt then what'll happen is that their internal states will somehow start to correlate strongly in in different ways and in different patterns. and perhaps with a temporality to where it's not as simple as drawing a singular um line but rather it'll be something much more complex like the dynamics for example of a Laurens attractor whenever we we draw it out and how the state space looks from there. Um so all this is to say um much of this chapter while at first glance provides us with notation and the necessary equations for solving for minimizing free energy in various ways and continuous state spaces uh using those kinds of models. We're also along the way that the author authors cleverly snuck in all of these other aspects regarding synchronization, communication, some of the building blocks for thinking about multi- aent simulations. What does it mean for two systems, whether it be an agent and environment or it be two agents or otherwise? How do how do they synchronize one another? And then we're also reintroduced to the POMDP structure which we see is like everything in the upper half of this graph excuse me it is graph in figure 8.6 six um basing graph and then the the upper half is the PMDP. Everything below around where this dash line is we can see is a lower level uh faster time scale hybrid model in the sense that for every iteration of the POMDP up here we see that there are actually three iterations of the continuous state space model at the lower level. So it's kind of like you can imagine receiving a lot of data over time and then at some threshold of time steps such as three in this case we could send that upward to the higher level to say okay we receive this data what does it mean right um it's kind of like taking in the full experience and then from there the poomdp would then use its own model parameters to update the different kinds of inferences to send those back down as priors to level one as we continue the inference process. Um I and we we have some nice discussion again looking at sakads and um those kinds of dynamics which are much more clearly continuous state spaces more often than not when looking at kind of like continuous coordinate systems. Uh we also have mixture models and clustering which again kind of uh imagine for those unfamiliar with that kind of logic imagine taking that hybrid model we just saw and kind of smashing it down such that we can receive um continuous data but then use it to infer clusters or discrete state space data. Right? So you you see a certain kind of pattern in the physiology and you deter you you determine that someone shows a particular discrete health state such as they are healthy versus another combination of continuous values their heart rate their glucose levels and so on. And in this particular state space uh discrete state space space they are they are um moderately unhealthy right and and with this set of values right so it's kind of going continuous to discreet um so I think I'll leave it there because I' I've already talked for quite some time apologies for that if it took away from anyone um I'd like to open up the conversation then if anyone has any questions or anything in particular particular that stuck out to them in this chapter. And we can discuss it if you like. There's always the um I'll make a first comment and then John or Isabella like in the first equation of the chapter it uh But you said neural signals are measured continuously. So often times people look at the discrete spiking of a single neuron you know even though that's that's a simplification but measurements are often continuous and then later discretized. So it's sort of like if you look at the instrumentation that's actually coming off of real world sensors in a way they're kind of like analog and then digital systems discretize that for the for for different kinds of advantages but um analog measurements are prevalent and then this is just the the um even though confusingly with the G and an F that don't map to equation 2.6 six and 2.5. This is sort of a different look on the S and the O from the POMDP in terms of having uh observables Y which is O latent states X which is S and then instead of having the B transition matrix interpolated between different latent states rather there's a continuous um first derivative landscape with X dot so kind embodies the same uh or analogous elements but with different letters. Yeah, thanks Daniel. Yeah, it's good points and because it's worth just really getting familiar more with the notation for anyone who's still kind of picking up on active inference and I with this in one way and I don't I don't want to these are my words not the author's words of course but in a sense if you think about the the A matrix as a in the in the discrete state space case in chapter 7 um and we think of how that is relating hidden states to observations. Then we do have something like a relationship here for this generative model that very much relates the two. And then if we think of a B matrix as the probability of the next state conditioned on the current state and uh the current action or policy being pursued. Um this of course is different. We're looking at derivatives or rates of change. But nonetheless, we are essentially in this first example, we're relating hidden states and observations. In the second, we are relating hidden states to something like the temporal aspect of hidden states where the hidden state will come next. So of course, it's not a transitioning to a a new state necessarily, but it's looking at its rate of change, which then of course could be used to predict what the value of the next hidden state will be in this continuous state space. And then we also have you know V almost like its own third variable and it itself doesn't have to be static. This of course is just like a nice simple example and it can be used as an attractor in this case. And then also with neuronal signals just with my own some of my own previous work on uh EEG signals it's so interesting in the sense of like a lot of them still are being uh recorded these signals are being recorded like digitally uh increasingly so and so you're kind of limited to the kind of rate at which the technology records and so even though you might looking at something down to the tenth of a millisecond. If we think about how when we take a derivative, we're technically like having to do a lot of um discretization of of values in order to you know uh carry out differentiation or or integration. Then then similarly with EEG it's like okay well I'm still discretized to the tenth of a of a nancond or or a of a millisecond or whatever you know rate it's recording at. So it is it is interesting like once you get down to these fine fine lines it's like what is discreet or continuous any anymore that's much more dramatic than than probably what the sentence that should actually come to mind but still it's uh there's more nuance but but still it's um it is interesting to think about and for all intents and purposes for the most part with with neurohysiological or any kind of physiological data you still are typically looking at continuous data unless unless there's some kind of particular criteria, right? That that that that someone makes a judgment about a value they see and then they just mark it as like, you know, on a liyker scale 1 to5. It's still a discrete state space with an implied ordering to it. Uh for example, so anyway, but now we're talking a lot about a I'm talking a lot about data types now. Um, so see maybe something we could pick up on in the chapter 8 questions unless anyone No. Okay, no problem. Yeah, let's look at chapter 8 of the KOD. Nice summary that Daniel and Ally did a couple years ago. Let's see. Yeah. Another random thought is it's just interesting that the analytical relationships hold for the discrete case at all when they're more underlying differentiable flow-based form is continuous. And yet it works like almost exactly how one would think in the discretized setting kind of however you discretise it. So it's like underlying circle and then whatever resolution your screen is. It's like it kind of works out. But that doesn't seem guaranteed by the definition of a circle. But yet it is. I know it's an interesting way to put it. It's I mean it it's funny. It sometimes it comes down to like making your own kind of decision whenever it comes to the modeling like do you want to view something as discreet? But then of course if you're going to build an active inference like model then it becomes insanely important to decide which state space you're working with right because we then we do start to see you know I mean underneath there's similar mechanics between the two kinds of models if we're able to just say oh there's just two kinds there continuous and discreet with all the infinite varieties in between I suppose but um but still we end up with with very different dynamics and also worth talking about then um I mean I don't know no one's had any questions about it so I haven't gone into it but there are the generalized coordinates of motion which might just be but these are essentially sort of the the actual mathematical mechanics behind continuous state space models. Right? So it ends up what what ends up happening is that for those who are familiar with Taylor series approximation like you can view it as just you know you over you start to expand the number of of terms in your t uh tailaylor series expansion. Um the the authors note that around six generalized coordinates are sufficient, meaning we're going to expand that far. Um and it works quite well not just as supplying these core functions and equations. And then furthermore, they're also very commonly used in analyzing neuronal signals. Um but many obviously many many many other types of data that don't have to have anything to do with neuronal anything necessarily but these are classic equations that you would find there and then it also works quite well in the sense of having a variational inference model right because as opposed to trying to like excessively predict into the future what will happen we can kind of simplify to or approximate a behavior often times as authors say with six generalized coordinates. Let's look at yeah sensory attenuation. What are the causes and consequences of sensory attenuation? Just going to check the chat. My understanding nothing more than stochasticity within the agent. Chaotic movement from one attractor or internal state to another. Movement and sensory attenuation. A point where you don't want to perfectly or accurately predict sensory consequences or associates of movement. Box 8.1. Belief I am not moving. Need to downweight the precision of that belief. Make it wider or lighter so that action shift can occur. policy shift occurs. Increase the attention again. Wow, we're somewhere between uh a sentence and a set of short causal sort of predicates here. Uh very symbolic, but actually we were getting warmer over time as we continued reading down these statements. This is not a bad way of compressing information that could help us to describe sensory attenuation. Why is that sensory attenuation? So, one of the interesting things that can sometimes go overlooked in active inference and it's worthwhile to to know be aware of it for anyone who's interested in uh you know kind of understanding and then utilizing the active inference framework. We know that agents are self-evidencing, right? So self-evidencing. Well, that means that in some way, if we just take it at face value, the idea is that an agent will try to realize what it already expects, right? We saw this in chapter 7 with preferences. And the more expanded, you know, way of looking at them is prior over observations. And if they remain fixed, which often times they do, such as in the T-mas, that the mouse always wants the cheese. At no point does it start to want the shock and devalue the cheese, right? So that reminds us of the kind of um attractor dynamics, right? You know, and so sometimes people have used the phrase goal prior, which probably should not be too terribly confused with the phrase pref preferences, but it has been used in in place of it to say that this is a prior belief you have. And because it doesn't change, that is where you are attracted to, let's say. So whenever it comes to sensory attenuation, the odd thing comes up where in the moment you believe you are sitting still and you decide you would like to get up, right? But you can't because you strongly believe currently that you're sitting still, but maybe you have in your goal prior that you do or your preferences you do want to stand up. But you can imagine this kind of strange conflict that occurs even though presumably all of you wants in a more colloquial sense to stand up. So what is this sensory attenuation about and what is belief I am not moving need to downweight the precision of that belief so that action shift can occur policy shift occurs increase the attention again let's go to box 8.1 see of course it's a text precision control is its role in movement generation. To understand this, it is worth thinking about what happens in the absence of this control. Imagine first that sensory data are predicted with high precision. I am sitting down. I notice I'm sitting down. The messages from these data are therefore afforded high synaptic gain and lead to various inferences about the position of some body part. I am definitely sitting down. The problem with this is the equivalence between motor commands and predictions under active inference. An accurate belief that I am not moving. I'm not moving, I'm sitting down. Cannot be used to predict the sensory consequences of movement vital for the initiation of that movement. How will I ever move if I firmly believe I'm sitting down? The idea is to attenuate or or or sort of partially if not fully, I don't want to say fully, but but to some significant actionable degree to reduce one's precision in their beliefs. meaning that that that dynamic I described earlier of sort of a a flattening which is I they're implying that in order to stand up I need to actually temporarily attenuate my belief that I'm sitting down just so that the the proprioceptive signals and everything else can play out such that I can stand up right so so that there's this kind of interesting philosophical nearly aspect of this that is we have to convince ourselves implicitly. No assumption that any of this is actually explicit in the sense of fully conscious you're thinking this out loud in sentences or otherwise. But the sense is that you actually have to kind of pause your beliefs in some way or more literally it would be you have to introduce uncertainty in your understanding of the current situation in order to actually change that situation which is rather fascinating and it it it comes up is so it's something to potentially wrap your head around. It's but it it starts to make sense and then it becomes useful in a variety of situations, right? For example, um it's it's been found to be rather significant in cases of Parkinson's disease where someone is struggling to attenuate um their their sensory intake, right? they're they're they struggle to to have fuller control over how their body responds, of course. Uh and and so it's a kind of by being able to understand sensory attenuation and its connection to things like precision precision control as well as being able to see in other areas of the textbook how they relate precision and in particular precision values to potentially being relatable to various neurotransmitters. suddenly being able to understand these kinds of dynamics at least as of this point in time in the textbook a couple years ago by these authors and the work has not stopped since so and I don't think it's going to but at least as far as computational psychiatry is concerned being able to make these kinds of connections that are interpretable and have neurobiological substrates involved um can at least hold the promise of being able to better serve different kinds of interventions and treatments for those who do struggle with these kinds of pathological disorders uh in ways that were not necessarily visible before because of a lack of the kind of underlying computational dynamics. And that's sort of why fields like computational psychiatry have been sort of taking off the past several years as the breadth of computational tools ever widens and more and more people produce open-source resources so that broader populations of people can get involved in these kinds of work. And of course they're not purely limited to computational psychiatry whatsoever. We could imagine anyone who's working in robotics who's working with a you know uh some kind of robot whether it be a automation system that's building car parts or it be like an actual robot that has limbs and navigates the world on its own. In either case, imagine if it's running on inference principles, especially a basian one where we can have the opportunity to have excessively strong priors such that the robot never does anything because it's too terribly convinced of the current situation that it doesn't do anything to change it. Well, then suddenly sensory attenuation becomes one of the first concepts that you should look at. And it brings up interesting implications for neurotransmitters as well and relating them to other kinds of cognitive systems, right? Um being able to have the capacity for the synaptic gain precision control dynamics has a lot of implications for for building any kind of predictive system. Whenever we're looking at this or similar frameworks, what is getting out of a chair? I was not the first. Um, yeah, it's also a sensory attenuation. It's like especially when we think about awareness or higher level introspection, we're attenuating almost all perception and only with really dedicated focus on sensory input can we even attend to you just all of our skin all of the sounds all these other things all spatial awareness ability to think of any concept it's like it's all being attenuated away that's sort of the attention question so it's like there's the attention is all you need and then there's the other side of the coin point is attenuation. But that's the same thing. It's just like whether you focus on upweing what is to be focused on or downweing what isn't. Yeah. Yeah, definitely. I mean, it's like if I'm driving on the highway in Chicago, as I occasionally do, um, I mean, if I attempt to maintain attention, uh, whenever it comes to every single car that's currently in my frame of vision, I'm certainly going to have a tougher time focusing on the particular things I need to attend to, such as the car immediately in front of me. that is suddenly breaking and so I should respond. And so it's just another way of kind of reating Daniel's point that is the kind of attentional dynamics needed to optimally or at least suboptimally but enough to survive a situation let's say um like those are there and we can't really do without them. um so much of the sensory data that we have hypothetically available to us goes very unnoticed, right? So, and even for in the case of robotics like if you have a robot who is you know putting together sequentially different parts on a manufacturing belt um it's not if it takes in every single signal that it receives it might not be able to hone its attention in on the very particular task at hand. Right? So, um relates physiologically. We can think of things like metabolism. We can think about energy uh expenditure and and saving um or conservation. So if you took in every single thing you ever saw in its full entirety as if we had perfect, you know, um um like picture book like memory um photographic memory. There it is. And I mean thinking about it computationally, I mean that's an absurd amount of data. It's going to be a lot and if it all gets con, you know, um stored in long-term memory. So it's just it's it's such an it's interesting thing about I mean I've always appreciated in this chapter that despite the fact that it's strongly centered on in name continuous time models we actually sneak in a lot of really interesting concepts that are much harder to demonstrate or describe until we reach this part of the textbook. This is the um this is where they apply or look at luck of voltera dynamics again but instead we're looking at it in terms of this eyelink conditioning problem related to the neurobiological substrate underneath. And we're looking at these kind of classic ways of just looking at conditioned stimuli, unconditional stimuli. For those who've have backgrounds in psychology, this will be um almost annoyingly familiar. Um, we can look at how expectations play out, but then we also see these kind of potentially cyclic dynamics that play out that necessarily have to be that way, right? Or close enough to being that way such that the system can persist. And then over here, this was, you know, just as they're simulating the system of handwriting. This is kind of the way it plays out. And then it itself has a kind of near cyclical dynamic. Like we were to segment this, right? We see a nearly identical structures right next to each other in what from far away looks like a simple accidental scribble. [Music] This is a little bit different. to just do this. But so I notice still haven't had any particular questions and this is a Julia library called RX Infern. And there's been a nice partnership between the developers of RX and fur and the institute for quite some time now. Um a rather active group has formed around it uh for anyone who is ever interested and we're looking essentially at applying and learning and understanding as well as giving feedback to the developers for this package. Um, and I the reason why I bring it up here and now is that in my opinion, this offers some of the most interesting tools for looking at continuous time and such a cool website, right? Um, continuous time models. And so for anyone who wanted to get into this chapter but are still becoming more familiar with active inference itself and are really just wanting to look at how to apply active inference in ways that they're already familiar with maybe computationally. This website alone will show you a variety of ways to potentially accomplish what you want to accomplish. ranging from the various equations and how they draw out variational free energy as well as Beth approximation um talking more about the underlying dynamics involved that we get a very good picture of them in the textbook but very significant on this website as well plus you get to kind of see it like in action being tailored for direct modeling. So they have an entire set of examples that it looks like they're still doing some modifications to their website but it's all here. So we have basic examples basian linear regression just being able to simply look at the relationship between fuel consumption and speed. Faster you go the more fuel you burn.

hidden Markov model. They've been expanding their set of use cases that arcs infer could be used for. And so now we're suddenly looking at a hidden markoff model as we saw in chapter 7. This is particularly a Roomba navigating different parts of a house. And it the hidden markoff aspect is it relates its its visual sensory receptor of what the floor looks like to a hidden state. That is its belief about what room is it it is in over time in our classic perception loop. This is only perception uh hidden markov models. And then they're starting to increase even further lately. support for POMDP models. um which is great and suddenly makes this library incredibly relevant because that would mean that it also covers the kinds of state space models that are covered in Briston's classic SPM 12 package for mat lab as well as PIMDP for Python which um the SPM does definitely allow for different kinds continuous state space dynamic analysis but RX infer out of the box is very much ready to go for creating all kinds of models. Here's the Gaussian mixture model I mentioned earlier where you can take in continuous values Y and then use them to infer discrete clusterings, right? that are close enough together such that we decide, oh, this makes up a distinct cluster. Right? So, it's so it's just interesting to think about the the many many ways for for anyone who's familiar with deep learning. Some of this will just be second nature to already be thinking about these things. But whenever we think about active inference, we think about basian perspectives where we're always coming into a situation with a prior and thinking about different data types. And for anyone who's actually writing code and and producing models, what data they have available in the potentially infinite ways that they could put things together, not just what data are you using and and and what is the breadth of the types, but also what are the inference processes between them and what kind of state space are they in? Are you going to discrete discretise something yourself because it makes sense to you, but maybe it won't to someone else who looks at the same model? They wouldn't approach it that way. Auto reggressive models lagging a variable on itself. I mean these are classic models whenever it comes to looking at forecasting and looking at like you know where where data is sparse and then suddenly you decide oh if I use the same data stream but just lag it on itself maybe it is a function of itself like it suddenly the the single data source you have becomes useful for predicting itself in that sense so then we can see that play out with observations and then the hidden state inference kind of drags or follows after that. So they have this kind of lag that occurs over time. So of course with genera um with continuous state spaces you know as opposed to feeling where like we're constricted to the ab c and ease of pomdp models here you could have I mean of course your matrices in the pomdps could be massive in the sense of you can include a broad variety of of sensory receptors or or or data streams coming in. You could have a plethora of different kinds of hidden states or hidden state factors uh each with their own discrete hidden states within. You could make an incredibly complex model. But but that said with with continuous time models it often feels a little bit more general in the sense of you could have m you could have various parameters some for the same thing some for different things some which are highly related to one another but computed a bit differently and then at the end of the day you're when you're learning them it's just to somewhat follow the logic of a deep neural network or otherwise we're just looking at something Like how do we compute weights? But how do we do it with respect to minimizing variational free energy as a functional of those parameters and the data coming in? And furthermore, much more distinctly, how do we look at the rate of change in those systems? And then here was the nice generalized synchrony piece. How do we synchronize with one another in our daily activities? I say as someone with a social sciences background. So, um, great point with the neural network analogy and reinforcement and the reward function as needed to give a gradient descendable strategy to optimize a functional of weights which don't directly correspond to the semantics of the probability distributions themselves. It's just hope that there's enough of these tiny guys firing that you can recapitulate at a higher level the semantics of a target distribution versus doing variational free energy as a direct gradient tractable landscape on the probability distributions of interest and how learning can have a and then that learning rate is fixed or you could learn the learning rate and then how fast you learned that. It's like the buck stops somewhere in a given implementation even though there are like infinite dimensional approximations. It's what brings me back to the just the general sense of an attractor is well all these things could move. We know from chapter six that a lot of it as well as chapter nine I say as we wrap up here but uh chapter nine we also had that that classic you know we're we're I'll just say we're we're we're always making our own decisions of how things should play out and for one person learning a learning rate is incredibly exciting and a means of testing a very particular hypothesis they have and then for another person developing another kind of system or not developing a system rather just trying to create a generative model that generally accurately approximately approximates uh an organism of interest that they don't want to make any arbitrary decisions. They want everything to be as synced up to the organism they're looking at as possible. The idea of learning a learning rate becomes like dangerous. Is it going to throw your model results so far off or make them so terribly uh non-geneneralizable that it's like why do any of this in the first place or can you fit the model to find the learning rate you know and that that would be probably a more balanced perspective. Um but in any case thanks everyone for joining today. uh really appreciate it. And so a week from today, this same time will be cohort 8, we'll be talking about chapter three, I believe, the high road of active inference. And then two weeks from today at this time we'll move on to chapter 9 as I was just mentioning which we'll get much more into things like model fitting and different kinds of analysis and group analysis and wrap up with models of false inference. And that's very much where the kind of computational psychiatry line goes. But many others, can a robot be pathological? That kind of question phrased exactly like that is ever more strongly on the minds of many today. Um, so interesting ways of thinking about things, but um, yep, that's all for today, folks. So, thank you very much. Thank you, Andrew. Thanks, Isabella and John. Bye.